



Customer Shopping Behavior Analysis

End-to-end data analytics project uncovering revenue drivers, customer segments, and actionable business insights from ~3,900 customer records.

Key Business Questions

1

Revenue Drivers

Which customer segments drive the most revenue?

2

Discount Patterns

How does discount usage vary across product categories?

3

Subscription Value

Do subscribed customers spend more?

4

Shipping Impact

How does shipping type relate to average order value?

The Dataset



SOURCE DATA

customer_shopping_behavior.csv

~3,900 customer records · Synthetic retail dataset

- Demographics: Age, Gender, Location
- Transactions: Purchase Amount, Category, Item
- Behavior: Subscription Status, Discount Applied
- Logistics: Shipping Type, Payment Method
- History: Previous Purchases, Frequency



Tools & Technologies

Python

Pandas, NumPy, Matplotlib, Seaborn

SQL

PostgreSQL / MySQL / SQL Server

Power BI

Interactive dashboard & visualization

Reporting

Word / Google Docs + Gamma

Step 1 — Python EDA



What We Did

- Inspected shape, dtypes, null values
- Cleaned missing values & duplicates
- Visualized distributions & correlations
- Engineered `age_group` & `customer_segment` features

Notebook: `Customer_Shopping_Behavior_Analysis.ipynb`

Step 2 — SQL Business Queries



Revenue by Gender

Male vs. female total spend comparison



Shipping vs. Avg Spend

Standard vs. Express order values



Subscription Revenue

Revenue & avg spend by subscription status



Customer Segmentation

New / Returning / Loyal via CTE + Window Functions

Step 3 — Power BI Dashboard



Dashboard Features

- Revenue by gender, age group & season
- Subscription vs. non-subscription comparisons
- Discount usage by product category
- Shipping type performance metrics
- Customer segmentation breakdown

 File: `customer_behavior-dashboard.pbix`



Key Results & Insights

Subscribed Customers

Significantly higher revenue & avg spend — **loyalty programs drive value**

Loyal Customers (10+ purchases)

Smallest segment, yet **disproportionate revenue** — retention beats acquisition

Clothing

Top-performing category with most repeat purchases — **core for promotions**

Express Shipping

Correlates with **higher average order values** — attracts high-intent buyers

More Key Insights

Primary Demographic

Age group **35-54** drives the highest total revenue

3,900

Customer Records

Discount Hotspots

Highest discount usage in **accessories & footwear** — category-specific strategy needed

3

Segments

New · Returning · Loyal

How to Run the Project



Python Notebook

Install dependencies, launch jupyter notebook

Customer_Shopping_Behavior_Analysis.ipynb



SQL Queries

Import CSV as customer table, open customer_behavior_sql.sql in your SQL client



Power BI Dashboard

Open customer_behavior-dashboard.pbix in Power BI Desktop, refresh data source

✅ All files available in the customer-shopping-behavior-analysis repository.